

외계행성 대기

Exoplanetary Atmospheres

Lecture 3: Temperatures, Albedos, Flux Ratios, and Phase Curves 2026 September 21 (Monday)

Based on: Chapter 3 of Seager, *Exoplanet Atmospheres* (2010)

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Outline – Lecture 3

Energy Balance

Planetary Temperatures

Planetary Albedos

Planet-Star Flux Ratios

Planetary Phase Curves

Summary

From flux to temperature: the plan for Chapter 3

When observing a distant exoplanet, the only thing we can measure is the radiation coming from it: a *flux*. The quantity relevant for the physics and chemistry of the atmosphere is its *temperature*.

- ▶ Only a single flux may be measured, yet real planets rarely have a single temperature. Which temperature should represent the planet? Which temperature should represent the flux?
- ▶ Temperature and albedo are tied together by **energy balance**: what is not reflected is absorbed and must be reradiated.
- ▶ Planet-to-star flux ratios (next lecture) follow from the temperatures and albedos, and decide what kind and size of telescope is needed to detect a given planet.

This section: energy balance $\rightarrow$ three planetary temperatures (T_{eff} , T_{eq} , T_{b}) $\rightarrow$ four albedos (single-scattering, geometric, spherical, Bond) and the relations among them.

Energy balance

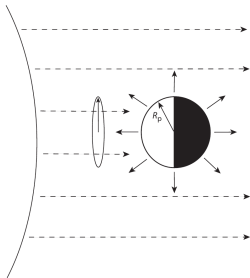


Figure 1: Planetary energy balance. Stellar radiation falls on the planet's cross-section πR_p^2 ; a fraction $(1 - A_B)$ is absorbed, here advected around the planet and reemitted uniformly into 4π . Reproduced from Seager (2010), Fig. 3.1.

No energy is created or destroyed in the atmosphere. All the energy comes either from the star (absorbed incident radiation) or from the planetary interior:

$$E_{\text{out}}(t) = (1 - A_B) E_{\text{inc}}(t) + E_{\text{int}}(t)$$

- ▶ E_{out} : energy per unit time leaving the planet.
- ▶ E_{inc} : stellar energy per unit time incident on the planet.
- ▶ $(1 - A_B)$: fraction absorbed, heating the atmosphere or surface, later reemitted at longer wavelengths.
- ▶ A_B : the **Bond albedo**, the fraction of incident stellar energy scattered back into space.
- ▶ E_{int} : energy per unit time supplied by the interior.

Energy balance in terms of fluxes

Write the outgoing energy as the planetary luminosity $L_p = 4\pi R_p^2 F_{S,p}$ and the incident energy with the result of Chapter 2, $E_{\text{inc}} = \pi R_p^2 (R_*/a)^2 F_{S,*}$:

$$4\pi R_p^2 F_{S,p} = (1 - A_B) F_{S,*} \left(\frac{R_*}{a} \right)^2 \pi R_p^2 + L_{p,\text{int}}$$

Two energy sources for the atmosphere: absorbed (and reradiated) starlight, and the planet's own interior luminosity $L_{p,\text{int}}$, the flux passing through a surface enclosing the planet from the inside.

- ▶ **Giant planets** (H/He): $L_{p,\text{int}}$ is the gradual loss of residual gravitational potential energy from the planet's formation (Kelvin–Helmholtz contraction, plus helium rain in Saturn). Jupiter's internal luminosity is more than *twice* its reradiated absorbed sunlight — an indication of how long it takes energy to travel from a giant planet's interior out to space.
- ▶ **Earth**: $L_{p,\text{int}}$ ($\simeq 44$ TW) comes partly from residual accretion energy but mostly from the decay of radioactive isotopes of uranium, thorium, and potassium. It is $\sim 2 \times 10^{-4}$ of the absorbed sunlight (1.2×10^{17} W, exercise 3.1b): negligible for the surface temperature, but it drives plate tectonics and volcanism, and thereby the long-term atmospheric evolution.

When can the interior be neglected?

For many planets the luminosity from external heating overwhelms the interior luminosity by many orders of magnitude:

- ▶ **Hot Jupiters** in very short-period orbits ($P < 4$ d): with $T_{\text{eq}} = 1600$ K and a Jupiter-like interior heat flow ($T_{\text{eff,int}} \simeq 120$ K), the ratio of interior to absorbed stellar energy is $(120/1600)^4 \simeq 3 \times 10^{-5}$ (exercise 3.1a).
- ▶ **Terrestrial planets**, with their typically limited interior energy (Earth: 2×10^{-4}).
- ▶ **Exceptions:** young giant planets and brown dwarfs (directly imaged planets are seen *because* of their interior luminosity), and the Solar System giants, where the interior contributes 30–100% of the emitted power.

We therefore usually write the energy balance without the interior term,

$$4\pi R_p^2 F_{S,p} = (1 - A_B) F_{S,*} \left(\frac{R_*}{a} \right)^2 \pi R_p^2$$

Time dependence and eccentric orbits

- ▶ The planet exists in an equilibrium with the incident stellar radiation in which the heating is constant in time, so from now on we drop the time dependence of the planet flux and related quantities. (Energy per unit time is technically called *power*.)
- ▶ Planets on **eccentric orbits** do *not* have constant heating: the incident flux varies as $1/r^2$ between periastron and apastron, by a factor $[(1+e)/(1-e)]^2$, which is 2.3 for $e = 0.2$ and 1000 for HD 80606b ($e = 0.93$).
- ▶ Exercise 3.3: replacing a by the instantaneous distance r and averaging the incident flux over the orbit gives the orbit-averaged equilibrium temperature $\langle T_{\text{eq}} \rangle = T_{\text{eq}}(a) (1 - e^2)^{-1/8}$, a weak correction because the planet spends most of its time near apastron.
- ▶ Whether the atmosphere follows the instantaneous heating or the average depends on the **radiative timescale** of the photosphere compared with the orbital period: hot, low-pressure photospheres respond within hours (HD 80606b's $8 \mu\text{m}$ flux rises by a factor of 5 within 6 hours of periastron passage), whereas deep, cold layers average over many orbits.

Three temperatures for a planet

There is no single temperature that describes a planet atmosphere: it varies with altitude, with horizontal position, and in time (day/night, seasons). Yet temperature governs, to first order, the chemical equilibrium state and hence the emergent spectrum. Three temperatures are in common use:

- ▶ **Effective temperature** T_{eff} — a proxy for the global temperature; defined from the *total* emitted flux. In principle a measured quantity.
- ▶ **Equilibrium temperature** T_{eq} — the T_{eff} a planet with no internal luminosity would have in equilibrium with its star. A *theoretical* number computed from R_* , $T_{\text{eff},*}$, a , and A_{B} .
- ▶ **Brightness temperature** $T_{\text{b}}(\nu)$ — the temperature of a blackbody with the same flux *at a given frequency*. The only temperature we can currently *measure* for exoplanets.

All three are defined by comparing the planet to a blackbody of the same size and distance; they differ in *which* flux is matched (total or monochromatic, measured or theoretical).

The effective temperature T_{eff}

T_{eff} is the temperature of a blackbody of the same shape and at the same distance as the planet with the *same total flux* (integrated over all frequencies). Using the blackbody flux πB and assuming a uniform surface flux,

$$F_S \equiv \int_0^\infty F_S(\nu) d\nu = \pi \int_0^\infty B(T, \nu) d\nu \equiv \sigma_R T_{\text{eff}}^4.$$

Carrying out the integral (exercise 3.2, using $\int_0^\infty u^3/(e^u - 1) du = \pi^4/15$) gives the **Stefan–Boltzmann law** and constant

$$F_S = \sigma_R T_{\text{eff}}^4, \quad \sigma_R = \frac{2\pi^5 k_B^4}{15 c^2 h^3} = 5.670 \times 10^{-8} \text{ J K}^{-4} \text{ m}^{-2} \text{ s}^{-1}$$

- ▶ σ_R is not a fundamental constant: it is built from h , c , and k_B . (Heng, problem 2.8.4: the same $\sigma_R T^4$ results whether one integrates B_λ , B_ν , or $B_{\tilde{\nu}}$.)
- ▶ In principle T_{eff} is a *measured* quantity: take the total flux of the planet, convert it to surface flux with $(D_\oplus/R_p)^2$, and find the temperature of the blackbody with the same total flux. In practice we never observe *all* wavelengths, so a model is needed to fill the gaps.
- ▶ F_S here is the planetary surface flux, in $\text{J m}^{-2} \text{ s}^{-1}$, radiated at the planet, as derived in Chapter 2.

What T_{eff} means physically

- ▶ T_{eff} describes the global temperature of the planet at the altitude where the *bulk of the radiation leaves* the planet or star. This altitude is sometimes called the planet's **photosphere** or “surface”, whether or not it is at the solid surface.
- ▶ T_{eff} may be very different from the surface temperature. **Venus:** $T_{\text{eff}} \simeq 230$ K, a value that comes from the cloud tops which obscure the solid surface. Because of a strong greenhouse effect the surface temperature is 730 K, almost 500 K hotter than T_{eff} . **Earth:** $T_{\text{eff}} \simeq 255$ K versus a mean surface temperature of 288 K, a greenhouse warming of 33 K.
- ▶ T_{eff} varies widely among planets. The coldest planets of the Solar System, Uranus and Neptune, have $T_{\text{eff}} = 59$ K; Jupiter 124 K; Earth $\simeq 255$ K. If we could measure T_{eff} for exoplanets, we would expect the hottest hot Jupiters to have $T_{\text{eff}} > 2000$ K, and ultra-hot Jupiters such as KELT-9b exceed 4000 K on the dayside.
- ▶ Because T_{eff} is defined by the *total* flux, it says nothing about the spectral shape (next slide).

T_{eff} applied to two stars

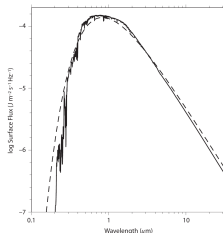


Figure 2: Surface flux of a $T_{\text{eff}} = 6130$ K G0IV star (HD 149026; solid) and the 6130 K blackbody (dashed). Seager (2010), Fig. 3.2.

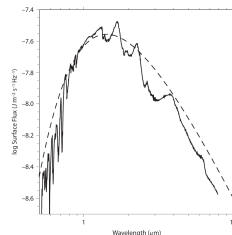


Figure 3: Same for a $T_{\text{eff}} = 3600$ K M2.5V star (GJ 436); the deep features are TiO (visible) and H₂O (near-IR). Seager (2010), Fig. 3.3.

Both stars have the same total flux as their blackbodies (equal areas under the curves). For the G star the spectrum is close to a blackbody; for the M star it departs strongly. *Lesson:* T_{eff} says nothing about the spectral shape. A planet around GJ 436 receives far more near-infrared and far less blue light than a blackbody would suggest, which matters for the Bond albedo (weighted by the stellar spectrum), for the photochemistry (little UV), and for the interpretation of secondary-eclipse depths (the stellar brightness temperature in the bands is not T_{eff}).

The equilibrium temperature T_{eq}

T_{eq} is an estimate of T_{eff} for a planet with *no internal luminosity*: the effective temperature attained by an isothermal planet in complete equilibrium with the radiation of its star. It refers to no flux measurement and is essential for exoplanets, for which T_{eff} is hard to measure. Start from energy balance, with a correction factor f to the 4π so that we can describe one hemisphere:

$$\frac{4\pi}{f} R_p^2 F_{S,p} = (1 - A_B) F_{S,*} \left(\frac{R_*}{a} \right)^2 \pi R_p^2.$$

Insert the Stefan–Boltzmann law $F_{S,p} = \sigma_R T_{\text{eq}}^4$ and $F_{S,*} = \sigma_R T_{\text{eff},*}^4$ and define $f' = f/4$:

$$T_{\text{eq}} = T_{\text{eff},*} \left(\frac{R_*}{a} \right)^{1/2} [f'(1 - A_B)]^{1/4}$$

- **Scaling:** $T_{\text{eq}} \propto a^{-1/2}$ and $T_{\text{eq}} \propto R_*^{1/2} T_{\text{eff},*}$; the albedo enters only as $(1 - A_B)^{1/4}$ (a factor 0.9 for $A_B = 0.3$).
- Since $L_* \propto R_*^2 T_{\text{eff},*}^4$, equivalently $T_{\text{eq}}^4 = f'(1 - A_B) L_* / (4\pi \sigma_R a^2)$: the temperature is set by the stellar flux at the planet's orbit.

The redistribution factor f'

For exoplanets we observe only one hemisphere at a time; this is what drives the need for f' . It describes *how the absorbed stellar energy is distributed* over the planet before being reradiated.

- ▶ **Uniform redistribution** ($f' = 1/4, f = 1$): the absorbed energy is spread over the whole sphere ($4\pi R_p^2$) and T_{eq} is the same for any hemispheric view. For Earth: $T_{\text{eq}} = 5778 \text{ K } (R_{\odot}/\text{AU})^{1/2} [0.7/4]^{1/4} = 255 \text{ K}$.
- ▶ **Instantaneous reradiation, no advection** ($f' = 2/3, f = 8/3$): a slowly rotating or tidally locked planet absorbs energy only on the dayside and each element reradiates locally. The dayside hemisphere is then much hotter: $(8/3)^{1/4} = 1.28$ times the uniform value. The value $2/3$ is the same number as the geometric albedo of a Lambert sphere, and arises from the same $\cos \vartheta$ -weighted hemispheric integral: the observer sees the hot substellar region face-on and the cooler limb obliquely.
- ▶ Between these two end cases there is no simple analytic form. To estimate the actual dayside temperature one needs observations (phase curves) or atmospheric circulation models (Chapter 10).
- ▶ **Caution:** f has many different definitions in the literature ($f = 1/4, 1/2, 2/3, \dots$ meaning different things). Seager's choice $f' = f/4$ as a correction to 4π is deliberate, for symmetry with the albedo derivation for scattered light from the dayside.

T_{eq} versus orbital distance

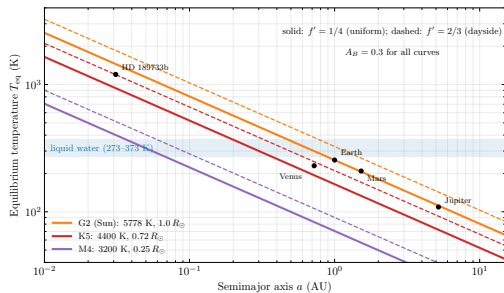


Figure 4: Equilibrium temperature versus semimajor axis for three host stars, with $A_B = 0.3$; solid $f' = 1/4$, dashed $f' = 2/3$. Points are Solar System T_{eq} values (Table 3.1) and HD 189733b.

- ▶ The $a^{-1/2}$ slope: moving a planet 100 times closer raises T_{eq} by a factor of 10 (Earth at 0.01 AU would be ~ 2500 K).
- ▶ Around an M4 dwarf the same T_{eq} is reached at $a \simeq 0.05$ AU: the habitable zone shrinks inward together with T_{eq} , which is why M-dwarf habitable-zone planets transit frequently and are tidally locked.
- ▶ HD 189733b ($a = 0.031$ AU, K dwarf): $T_{\text{eq}} \simeq 1200$ K for $f' = 1/4$, close to the measured dayside brightness temperature (Lecture 1), indicating efficient day–night heat transport.
- ▶ The dashed curves show how much hotter a non-redistributing dayside can be.

How good is T_{eq} ? The Solar System test

Planet	T_{eff} (K)	T_{eq} (K)	A_g (500 nm)	A_B
Mercury	—	—	0.106	0.119
Venus	~ 230	230	0.65	0.750
Earth	~ 255	253	0.367	0.306
Mars	~ 212	209	0.150	0.250
Jupiter	124.4 ± 0.3	109	0.52	0.343
Saturn	95.0 ± 0.4	80	0.47	0.342
Uranus	59.1 ± 0.3	58	0.51	0.300
Neptune	59.3 ± 0.8	46	0.42	0.290

Seager (2010), Table 3.1; values from Cox (2000). T_{eq} uses the measured A_B , the known a , and $f' = 1/4$.

- For the terrestrial planets T_{eff} and T_{eq} agree to within a few kelvin: their interior heat is negligible.
- For Jupiter, Saturn, and Neptune $T_{\text{eff}} > T_{\text{eq}}$ by 15–30%, the signature of their **internal luminosity** ($T_{\text{eff}}^4 = T_{\text{eq}}^4 + T_{\text{int}}^4$ with $T_{\text{int}} \simeq 100$ K for Jupiter). Uranus is the exception: $T_{\text{eff}} \simeq T_{\text{eq}}$, so its interior heat flow is anomalously small — still not fully understood.
- Note that $A_g \neq A_B$ in general, and that A_g can exceed A_B (Jupiter, Uranus, Neptune): the two albedos measure different things (next section).

The brightness temperature $T_b(\nu)$: motivation

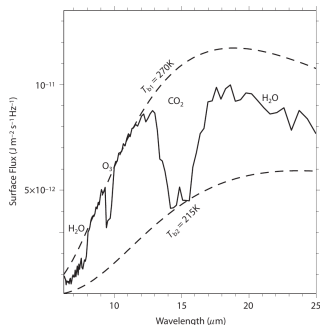


Figure 5: Brightness temperature. Solid: Earth's surface flux measured by Mars Global Surveyor. Dashed: blackbody fluxes at two temperatures. Reproduced from Seager (2010), Fig. 3.4.

The flux of exoplanets at different wavelengths or frequencies is the quantity we can measure at Earth, $F_{\oplus} = (R_p/D_{\oplus})^2 F_S$ for a very distant planet with surface flux F_S .

- Our intuitive familiarity with temperature makes it easier to use a temperature instead of a flux to describe exoplanet atmosphere measurements.
- Temperatures are more directly relevant than fluxes for the physics and chemistry of the atmosphere.
- Fluxes depend on the planet's distance from Earth and so change from one planet to the next, whereas the temperature of a planet atmosphere is distance independent.

Measured exoplanet fluxes are therefore often stated as **brightness temperatures**: the temperature of a blackbody of the same shape and at the same distance as the planet with the *same flux in a specified frequency range*.

The brightness temperature $T_b(\nu)$: definition

$$\pi B(T_b(\nu), \nu) = F_S(\nu) = F_{\oplus}(\nu) \left(\frac{D_{\oplus}}{R_p} \right)^2$$

- ▶ $T_b(\nu)$ is a frequency-dependent expression of the planetary flux: invert the Planck function at the observed frequency for the temperature. Seager writes this with the Stefan–Boltzmann form $\sigma_R T_b^4(\nu) = F_S(\nu)$ as shorthand for “convert flux to temperature”; the monochromatic Planck inversion is the operational definition used in practice.
- ▶ The brightness temperature assumes that the planet behaves as a blackbody *only* in the frequency range where the flux is measured; nothing is assumed about other frequencies.
- ▶ While a blackbody radiator has a constant $T_b(\nu)$, planets with spectra that depart from a blackbody have a $T_b(\nu)$ that can vary widely with wavelength or frequency.
- ▶ In the Rayleigh–Jeans regime T_b is simply proportional to the flux, $T_b = c^2 F_S(\nu) / (2\pi \nu^2 k_B)$, which is the definition used in radio astronomy; in the Wien regime a small change in flux corresponds to a small change in T_b , so brightness temperatures are a compressed, robust way to quote infrared fluxes.

Reading brightness temperatures

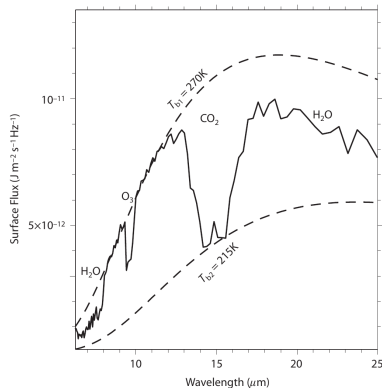


Figure 6: Earth's mid-infrared spectrum:
 $T_b(12\ \mu\text{m}) \simeq 270\ \text{K}$ but $T_b(14.2\ \mu\text{m}) \simeq 215\ \text{K}$.
Reproduced from Seager (2010), Fig. 3.4.

- ▶ For Earth: $T_b \simeq 270\ \text{K}$ in the $12\ \mu\text{m}$ window, where we see the warm surface through a transparent atmosphere, but $T_b \simeq 215\ \text{K}$ in the $14\text{--}15\ \mu\text{m}$ CO_2 band, where the atmosphere is opaque and we see the cold tropopause region. The $9.6\ \mu\text{m}$ O_3 band gives an intermediate value from the stratosphere.
- ▶ $T_b(\nu)$ is the *only* temperature we can currently measure for exoplanets, because it requires flux at just one frequency or wavelength range: a secondary-eclipse depth in the Spitzer 3.6, 4.5, 5.8, and $8\ \mu\text{m}$ bands gives four brightness temperatures of the dayside.
- ▶ To get from a measured $T_b(\nu)$ to the more global T_{eff} we must resort to a model atmosphere code that fills in the unobserved wavelengths.

Relating the three temperatures

	definition	measured or theoretical?	typical use
T_{eff}	blackbody with the same <i>total</i> flux: $\sigma_{\text{R}} T_{\text{eff}}^4 = F_{\text{S}}$	measurable in principle (needs all λ)	global energy output; interior heat
T_{eq}	T_{eff} of a planet with no interior heat, from R_* , $T_{\text{eff},*}$, a , A_{B} , f'	theoretical	first estimate for any new planet
$T_{\text{b}}(\nu)$	blackbody with the same flux <i>at ν</i>	measured (one band suffices)	reporting eclipse depths; vertical structure via $T_{\text{b}}(\nu)$

- ▶ $T_{\text{eff}}^4 = T_{\text{eq}}^4 + T_{\text{int}}^4$ when interior heat matters (the two fluxes add); $T_{\text{eff}} \simeq T_{\text{eq}}$ otherwise, provided the right f' is used for the hemisphere observed.
- ▶ All three compare the planet with a blackbody of the same size and distance; they differ only in *which* flux is matched: total or monochromatic, measured or computed.

Temperatures of a tidally locked planet

- ▶ $T_b(\nu)$ scatters around T_{eff} : it is higher in spectral windows that see deep, warm layers and lower in strong absorption bands that see high, cold layers, for a temperature decreasing with altitude. A temperature *inversion* reverses this: the bands appear in *emission* and T_b is higher inside them. Thus $T_b(\nu)$ across a spectrum is a crude thermometer of the vertical structure, which Chapter 9 turns into a quantitative retrieval.
- ▶ For a tidally locked hot Jupiter the *dayside* T_b measured at secondary eclipse can exceed $T_{\text{eq}}(f' = 1/4)$ by up to $(8/3)^{1/4} = 1.28$ if the absorbed energy is not redistributed, whereas the *nightside* T_b measured near transit can be far below it. The day–night contrast, i.e. the effective f' , is a diagnostic of the winds (Chapter 10).
- ▶ Example: HD 189733b has $T_{\text{eq}}(f' = 1/4) \simeq 1200$ K; the measured $8\text{ }\mu\text{m}$ brightness temperatures are $\simeq 1210$ K on the dayside and $\simeq 970$ K on the nightside (Knutson et al. 2007), so heat redistribution is efficient. For WASP-18b, in contrast, the dayside is close to the $f' = 2/3$ limit and the nightside is very cold.
- ▶ A single measured T_b therefore does not determine A_B : with f' unknown, albedo and redistribution are degenerate in the dayside temperature. Breaking the degeneracy needs a phase curve or an optical eclipse depth (Lecture 3).

What makes a planet bright or dark?

The **albedo** is the ratio of the light scattered by a planet to the light received by it. As for temperature, there is no single albedo; but the albedo is an important parameter because the reflectivity is indicative of cloud or surface conditions, and because it controls the energy balance and hence T_{eff} .

- ▶ **Venus** is the brightest planet, scattering 0.75 of the incident energy: 100% coverage by H_2SO_4 clouds.
- ▶ **Earth** has highly reflective water/ice clouds but only $\sim 50\%$ cover, and scatters 0.3.
- ▶ **Icy bodies** are bright, fresh ice more than old ice. Enceladus, with water geysers producing fresh ice, is extremely bright.
- ▶ **Mercury**, with no atmosphere, clouds, or ice, scatters only 0.1. Hot Jupiters are similarly dark ($A_g \lesssim 0.1$): no bright cloud decks, and strong Na/K absorption.

Four albedos and their relations: **single-scattering** $\tilde{\omega}$ (one particle), **geometric** $A_g(\nu)$ (full phase, relative to a Lambert disk), **spherical** $A_S(\nu)$ (all directions, one frequency), **Bond** A_B (all directions, all frequencies) — plus the **apparent** albedo, the one that can be measured for exoplanets.

The single-scattering albedo $\tilde{\omega}$

- ▶ The single-scattering albedo $\tilde{\omega}$ is the fraction of incident light that is scattered (rather than absorbed) by a *single particle* in the atmosphere or on the surface. It is wavelength dependent.
- ▶ Examples: a water-ice crystal in Earth's atmosphere can reach $\tilde{\omega} \simeq 0.8$; a plant leaf at visible wavelengths is closer to 0.1. Old to fresh snow 0.45–0.85; clouds 0.35–0.8; desert sand 0.2–0.35; ocean water < 0.05 .
- ▶ $\tilde{\omega}$ is *not* a global albedo of the planet. Nevertheless, the reflective properties of the individual gas, cloud, and surface particles ultimately produce the planet's global albedo through complex *multiple* scattering of the incident radiation.
- ▶ Equally important is the *directional* scattering property of a particle. Few particles in nature, if any, scatter isotropically; ice crystals, for example, have a strong tendency to backscatter. For a surface this is described by the bidirectional reflectance distribution function (BRDF); for a single particle by the phase function $P(\Theta)$ (Chapter 5).
- ▶ In radiative-transfer language (Lecture 3), $\tilde{\omega} = \sigma_s / (\alpha + \sigma_s)$, the ratio of the scattering coefficient to the total extinction. Conservative scattering, $\tilde{\omega} = 1$, is the Lambert-sphere idealization; a Rayleigh-scattering atmosphere over a black surface has $\tilde{\omega} \rightarrow 1$ in the blue and $\rightarrow 0$ in the red.

The phase angle α

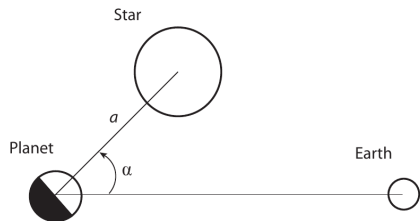


Figure 7: The phase angle α is the star–planet–observer angle; a is the semimajor axis. Reproduced from Seager (2010), Fig. 3.5.

Before defining the planetary albedos we need the **phase angle**: the star–planet–observer angle.

- ▶ $\alpha = 0^\circ$ corresponds to full phase: the observer sees the entire illuminated hemisphere.
- ▶ $\alpha = 90^\circ$ is quadrature (“half” phase); this is where the planet’s angular separation from the star is largest for a circular orbit, the natural configuration for direct imaging.
- ▶ $\alpha > 170^\circ$ corresponds to a thin crescent, and at $\alpha = 180^\circ$ the planet is not illuminated at all as seen by the observer (it is between the observer and the star: transit).

Phase angles in the Solar System and for exoplanets

- ▶ **Solar System:** only Mercury and Venus, interior to Earth's orbit, show all illumination phases from Earth; the outer giant planets show only a few to several degrees of phase angle. Full phase, $\alpha \simeq 0$, is the natural configuration for observing Solar System planets, with the Sun, Earth, and planet aligned at opposition.
- ▶ For exoplanets the range of visible planetary phases depends on the **orbital inclination** of the planet–star system.
- ▶ A planet in a system with $i = 90^\circ$ (edge-on) shows all phases as seen from Earth, except phases right near $\alpha = 0$ when the planet is directly behind the star (secondary eclipse) and near $\alpha = 180^\circ$ when it is in front (transit).
- ▶ In contrast, a planet with a face-on orbit ($i = 0^\circ$) has a constant “half” phase, $\alpha = 90^\circ$, throughout its orbit. In general the accessible range is $90^\circ - i \leq \alpha \leq 90^\circ + i$.

Full phase is never quite observable

- ▶ Because of the randomness of exoplanet orbital inclinations, $\alpha \approx 0$ may never occur for some planets.
- ▶ In reality $\alpha \equiv 0$ does not occur for any exoplanet, because the planet would be directly behind its star and not observable; however, we consider a small α close enough to $\alpha = 0$ for our purposes — for a transiting planet, the moments just before and after secondary eclipse, where $\alpha \simeq R_*/a \sim 5^\circ$.
- ▶ The phase angle is related to the orbital phase: for a circular edge-on orbit, $\cos \alpha = -\cos(2\pi\varphi)$ with φ the orbital phase measured from transit, so that α sweeps from 180° at transit through 90° at quadrature to 0 at eclipse. For inclination i , $\cos \alpha = -\sin i \cos(2\pi\varphi)$.
- ▶ For directly imaged planets the situation is reversed: small α (planet behind or in front of the star) is inaccessible because of the angular proximity to the star, and the accessible phases cluster around quadrature. Reflected-light imaging therefore measures the apparent albedo near $\alpha \simeq 90^\circ$ rather than A_g .

The geometric albedo $A_g(\nu)$: definition

The geometric albedo is the ratio of a planet's flux at *zero phase angle* (full phase) to the flux from a **Lambert disk** at the same distance and with the same cross-sectional area (i.e. subtending the same solid angle).

- ▶ The definition is historical: the brightness of a Solar System planet on a photograph was compared to that of a uniformly illuminated Lambert disk, a *relative* measurement that cancels instrumental errors.
- ▶ $A_g(\nu)$ is *not* the fraction of incident energy scattered at $\alpha = 0$ (exercise 3.5: that fraction is A_g/π).

Numerator: the planet flux at Earth at $\alpha = 0$, from Chapter 2 in latitude–longitude coordinates, keeping only *scattered* intensity (not thermal reemission):

$$F_{\oplus}(\nu) = \left(\frac{R_p}{D_{\oplus}} \right)^2 \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{S,\text{scat}}(\theta, \phi, \nu) \cos \phi \cos^2 \theta \, d\theta \, d\phi.$$

Denominator: a Lambert disk of radius R_p scatters isotropically with $I_{S,\text{scat}} = I_{\text{inc}}$, uniform over the disk, so

$$F_{\text{L disk}}(\nu) = \left(\frac{R_p}{D_{\oplus}} \right)^2 \pi I_{\text{inc}}(\nu).$$

The geometric albedo: general expression

Dividing numerator by denominator,

$$A_g(\nu) = \frac{\int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{S,\text{scat}}(\theta, \phi, \nu) \cos \phi \cos^2 \theta \, d\theta \, d\phi}{\pi I_{\text{inc}}(\nu)}$$

- ▶ Something interesting: the denominator $\pi I_{\text{inc}}(\nu) = F_{\text{inc}}(\nu)$ is the incident stellar flux at the *substellar point*. So A_g may also be described as the ratio of the planet's scattered flux to the incident stellar flux at the substellar point.
- ▶ This is the form to use to compute A_g from the output of a *model atmosphere code*, which delivers the scattered intensity at the planetary surface as a function of location, $I_{S,\text{scat}}(\theta, \phi, \nu)$.
- ▶ The subscript “scat” means that only *scattered* radiation is counted, not radiation that has been absorbed and thermally reradiated; at visible wavelengths, for planets cooler than ~ 1500 K, the distinction is clean.

The geometric albedo: making the physics explicit

For a more conceptual understanding, relate the scattered intensity to the incoming intensity with the dimensionless $p(\Theta)$: the fraction of incident intensity scattered out of the planet into the angle Θ from the incident direction. All the physics of directional and multiple scattering inside the atmosphere is buried in p (it is *not* the single-scattering phase function P of Chapter 5):

$$I_{S,\text{scat}}(\theta, \phi, \nu) = I_{\text{inc}}(\theta', \phi', \nu) p(\Theta),$$

primes for the direction of incidence, no primes after scattering. For uniform incident stellar intensity, diluted away from the substellar point by the projection (Fig. 2.3),

$$I_{\text{inc}}(\theta', \phi', \nu) = \hat{\mathbf{n}} \cdot \hat{\mathbf{k}} I_{\text{inc}}(\nu) = \cos \Theta I_{\text{inc}}(\nu) = \cos \theta \cos \phi I_{\text{inc}}(\nu),$$

so that $I_{S,\text{scat}} = \cos \theta \cos \phi I_{\text{inc}}(\nu) p(\theta, \phi, \theta', \phi')$. Then $I_{\text{inc}}(\nu)$ cancels between numerator and denominator:

$$A_g(\nu) = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} p(\theta, \phi, \theta', \phi') \cos^2 \phi \cos^3 \theta \, d\theta \, d\phi$$

One factor $\cos \phi \cos \theta$ comes from the oblique illumination, the other $\cos \phi \cos^2 \theta$ from the flux projection and the solid angle toward the observer.

A_g of a Lambert sphere, and why A_g can exceed 1

For a **Lambert sphere** the intensity is scattered equally in all directions, $p(\Theta) = 1$, and

$$A_g^{\text{Lambert}} = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} \cos^2 \phi \cos^3 \theta \, d\theta \, d\phi = \frac{1}{\pi} \cdot \frac{\pi}{2} \cdot \frac{4}{3} = \boxed{\frac{2}{3}}$$

- ▶ $A_g = 2/3 < 1$ even though a Lambert sphere scatters *all* incident radiation and absorbs none. One third of the light is scattered out of the line of sight: the limb regions are illuminated obliquely *and* seen obliquely, so they contribute less than the corresponding parts of a flat disk.
- ▶ By the same definition, A_g *can be greater than 1*: it compares the planet not to the incident radiation but to a perfectly diffusing disk. **Enceladus** has $A_g = 1.38$ at visible wavelengths (Verbiscer et al. 2007): fresh ice backscatters strongly, so at $\alpha = 0$ it returns more light toward the observer than a Lambert disk would. Rayleigh-scattering atmospheres also backscatter more than Lambert ($A_g = 3/4$ for a semi-infinite conservative Rayleigh atmosphere).
- ▶ Measurements of A_g have been elusive for exoplanets (upper limits from MOST, CoRoT, Kepler; a few detections such as Kepler-7b with $A_g \simeq 0.3$), but A_g remains the standard way to classify theoretical models.

The Bond albedo A_B and the spherical albedo $A_S(\nu)$

- ▶ A_B is the fraction of incident stellar *energy* scattered back into space, including radiation into *all* directions and at *all* frequencies. By definition $A_B \leq 1$.
- ▶ The **spherical albedo** $A_S(\nu)$ is the same quantity at a specific frequency. The Bond albedo is the spherical albedo averaged over frequency *weighted by the incident stellar spectrum*:

$$A_B = \frac{\int_0^\infty A_S(\nu) F_{S,*}(\nu) d\nu}{\int_0^\infty F_{S,*}(\nu) d\nu}$$

(Seager writes $A_B = \int A_S d\nu$ as shorthand and states that the weighting by the incident radiation is implied; Heng's Eq. 2.10 gives the weighted form explicitly.)

- ▶ Consequently both A_B and A_S depend on the *host star's spectrum*: the same planet has a different Bond albedo around a G star and around an M star, because an M star emits most of its light in the near-IR where the atmosphere may be dark.

To derive $A_S(\nu)$ we must account for the light scattered at *every* phase angle, not just at $\alpha = 0$. At each α only a portion of the hemisphere facing the observer is illuminated (next slide).

Scattered flux at phase angle α

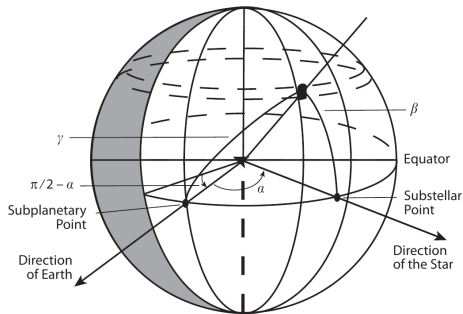


Figure 8: Coordinates for the phase-dependent scattered flux. θ is latitude from the intensity equator, ϕ longitude from the sub-Earth point, α the phase angle. After Sobolev (1975). Reproduced from Seager (2010), Fig. 3.6.

Measure longitude ϕ from the sub-Earth (subplanetary) point. The substellar point is then at longitude α , and a surface element at (θ, ϕ) is illuminated with the projection $\cos(\alpha - \phi) \cos \theta$ and seen with the projection $\cos \phi \cos \theta$. Only the longitudes $\alpha - \pi/2 < \phi < \pi/2$ are both lit and visible. The scattered flux toward the observer is

$$F_{S,\alpha}(\nu) = \int_{\alpha-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{\text{inc}} p \cos(\alpha - \phi) \cos \phi \cos^3 \theta \, d\theta \, d\phi,$$

and for uniform incident intensity we write

$$F_{S,\alpha}(\mathbf{v}) = I_{\text{inc}}(\mathbf{v}) \Psi_{\alpha}(\mathbf{v}),$$

where Ψ_α is the fraction of scattered radiation in the direction of the observer as a function of α . At $\alpha = 0$ it reduces to the geometric-albedo integral: $\Psi_{\alpha=0} = A_g/\pi$.

The spherical albedo: integrating over all directions

To count the energy scattered into all directions, surround the planet with a sphere and use α as the polar angle and β as the azimuth (azimuthal symmetry assumed). The total scattered energy per unit time per frequency is

$$E_{\text{scat}}(\nu) = I_{\text{inc}}(\nu) \int_0^{2\pi} \int_0^\pi \Psi_\alpha(\nu) R_p^2 \sin \alpha \, d\alpha \, d\beta = 2\pi R_p^2 I_{\text{inc}}(\nu) \int_0^\pi \Psi_\alpha(\nu) \sin \alpha \, d\alpha.$$

The incident energy (Chapter 2) is $E_{\text{inc}}(\nu) = \pi R_p^2 \cdot \pi I_{\text{inc}}(\nu)$. Their ratio is the spherical albedo:

$$A_S(\nu) = \frac{E_{\text{scat}}(\nu)}{E_{\text{inc}}(\nu)} = \frac{2}{\pi} \int_0^\pi \Psi_\alpha(\nu) \sin \alpha \, d\alpha$$

- ▶ A_S needs Ψ_α at *all* phase angles. From Earth we cannot see all phases of most planets, so A_S (and A_B) are not directly measurable: we need a way to estimate them from the one thing we can hope to measure near full phase, A_g .
- ▶ Note the different weighting from T_{eq} : the factor $2/\pi$ is $2\pi R_p^2 / (\pi R_p^2 \cdot \pi)$, the sphere over the disk and the π of the flux.

Relating A_S to A_g : the phase integral q

Using $\Psi_{\alpha=0}(v) = A_g(v)/\pi$, rewrite the spherical albedo as

$$A_S(v) = \frac{\pi A_g(v)}{\Psi_{\alpha=0}(v)} \cdot \frac{2}{\pi} \int_0^\pi \Psi_\alpha(v) \sin \alpha \, d\alpha \quad \Longrightarrow \quad \boxed{A_S(v) = A_g(v) q(v)}$$

with the **phase integral**

$$q(v) \equiv \frac{A_S(v)}{A_g(v)} = 2 \int_0^\pi \Phi_\alpha(v) \sin \alpha \, d\alpha, \quad \Phi_\alpha(v) \equiv \frac{\Psi_\alpha(v)}{\Psi_{\alpha=0}(v)}$$

where Φ_α is the **phase function**: the scattered flux versus phase angle normalized to full phase (this is the phase curve of the next lecture).

- ▶ A_S can be *estimated* from a measured A_g and a theoretical q . Heng writes the same thing as $A_S = A_g P_g$.
- ▶ **Lambert sphere:** $\Phi_\alpha = \frac{1}{\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha]$ (derived next lecture), so $q = 3/2$ and

$$A_g = \frac{2}{3} A_S, \quad A_S = A_B = 1, \quad A_g = \frac{2}{3}.$$

All light is scattered back to space, so the spherical albedo is 1; the geometric albedo counts only the light scattered toward the observer at $\alpha = 0$ and is less.

- ▶ **Rayleigh scattering:** $q = 4/3$; a Lambert *plane* (exercise 3.6): $\Phi_\alpha = \cos \alpha$. Real planets: $q \simeq 1.2$ – 1.7 .

Bond versus geometric albedo for the giant planets

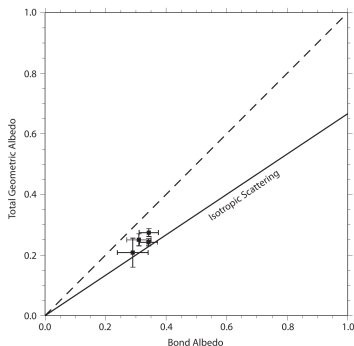


Figure 9: Total (wavelength-integrated) geometric albedo versus Bond albedo for a Lambert sphere (solid, $A_g/A_B = 2/3$) and for Uranus, Neptune, Saturn, Jupiter (points, in order of increasing A_B). The dashed line is $A_g = A_B$. Adapted from Rowe et al. (2006); reproduced from Seager (2010), Fig. 3.7.

- ▶ All four giants lie *above* the Lambert (isotropic scattering) line: their atmospheres backscatter more than a Lambert sphere, so $A_g/A_B > 2/3$ ($q < 3/2$).
- ▶ The wedge between the Lambert line and the line $A_g = A_B$ bounds the photometric properties of most spherical bodies with deep atmospheres (Rayleigh scattering, clouds, dust). A planet with deep atmosphere must lie to the right of the dashed line.
- ▶ Practical use: a measured (or upper-limit) A_g of an exoplanet, e.g. the MOST limit $A_g < 0.25$ for HD 209458b, can be converted to a range of A_B , hence to a range of T_{eq} .

The apparent albedo: the one an exoplanet can give us

The geometric, spherical, and Bond albedos are unsatisfactory for exoplanets because *none can be measured*:

- ▶ A_g is defined at full phase ($\alpha = 0$), when the planet is behind the star and unobservable (for transiting planets the eclipse depth measured just before/after eclipse is the practical substitute, at a small α).
- ▶ A_S and A_B require all phase angles. With direct imaging, phases where the planet projects too close to the star are inaccessible, and unless the orbit is edge-on the planet does not go through all phases at all.

Apparent albedo. The ratio of the scattered flux emerging from the planet toward the observer to the scattered flux from a *perfectly reflecting Lambert sphere* of the same size, at the same phase angle, at the same distance:

$$A_{\text{app}}(\alpha, \nu) = \frac{F_{S,\alpha}(\nu)}{F_{S,\alpha}^{\text{Lambert}}(\nu)} = \frac{A_g(\nu) \Phi_\alpha(\nu)}{\frac{2}{3} \Phi_\alpha^{\text{Lambert}}} \quad (\text{exercise 3.9}).$$

- ▶ It is the planet's reflectance in the direction of the observer at a given time, a function of phase angle and frequency.
- ▶ It is the only albedo that can be derived from a *single* observation of a planet of known size — a directly imaged planet at known separation and phase.

Temperatures and albedos: summary

Temperatures. Energy balance $4\pi R_p^2 F_{S,p} = (1 - A_B) F_{S,*} (R_*/a)^2 \pi R_p^2 + L_{\text{int}}$ gives $T_{\text{eq}} = T_{\text{eff},*} (R_*/a)^{1/2} [f'(1 - A_B)]^{1/4}$, with $f' = 1/4$ for uniform redistribution and $2/3$ for an instantaneously reradiating dayside. T_{eff} matches the *total* emitted flux and exceeds T_{eq} when the interior supplies heat; $T_b(\nu)$ matches the flux at one frequency and is the only temperature we can measure for an exoplanet.

Albedos. A_g compares the full-phase flux with that of a Lambert disk (a Lambert sphere has $A_g = 2/3$, and A_g can exceed 1). The spherical albedo is $A_S = A_g q$ with the phase integral $q = 2 \int_0^\pi \Phi_\alpha \sin \alpha d\alpha$ ($3/2$ for Lambert), and A_B is A_S averaged over the stellar spectrum — the one that enters the energy balance. Only the apparent albedo can be obtained from a single observation.

Next. With temperatures and albedos in hand we can write down the planet-to-star flux ratios that decide what is detectable.

Why flux ratios?

An estimate of the planet-to-star flux ratio tells us whether a given planet and star can be detected with a given telescope, at which wavelengths, and with what precision. Last lecture gave us the ingredients: surface fluxes, the flux at Earth, temperatures, and albedos. We now assemble them.

- ▶ Assume star and planet radiate as blackbodies (a first estimate; brightness temperatures refine it).
- ▶ A planet has **two spectral peaks**: a short-wavelength peak of *scattered starlight*, at the same wavelength as the star, and a long-wavelength peak of *thermal emission* at the planet's own temperature.
- ▶ During transit a third quantity appears: the *transmission* flux ratio, the fraction of starlight blocked by the atmospheric annulus.

The planet flux at Earth is $F_{\oplus,p} = F_{S,p}(R_p/D_{\oplus})^2$ and the stellar flux at Earth is $F_{\oplus,*} = F_{S,*}(R_*/D_{\oplus})^2$, so the distance cancels:

$$\frac{F_{\oplus,p}}{F_{\oplus,*}} = \frac{F_{S,p} R_p^2}{F_{S,*} R_*^2}$$

valid for total fluxes and for frequency-dependent fluxes alike.

Spectra of the Solar System seen from 33 light years

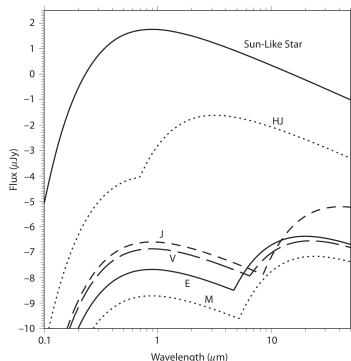


Figure 10: Approximate blackbody spectra (in $10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$) of the Sun (5750 K), Jupiter, Venus, Earth, Mars, and a representative hot Jupiter, seen from 33 light years (10 pc). Reproduced from Seager (2010), Fig. 3.8.

- ▶ Each planet shows the two peaks: reflected light near $0.5 \mu\text{m}$ (shape of the stellar spectrum, scaled by $A_g R_p^2 / a^2$) and thermal emission peaking at $\lambda_{\text{max}} \simeq 2898 \mu\text{m K} / T$: $\sim 10 \mu\text{m}$ for Earth and Venus, $\sim 25 \mu\text{m}$ for Jupiter, $\sim 2 \mu\text{m}$ for the hot Jupiter.
- ▶ The planet-star contrast is large at *all* wavelengths, but the thermal peak of a hot Jupiter reaches within 10^{-3} of the star, the Earth's thermal peak within 10^{-7} , while in the visible the contrast is 10^{-9} – 10^{-10} .
- ▶ Venus is brighter than Earth in reflected light (high albedo, closer to the Sun) but similar in the thermal infrared: its cloud tops radiate at $T_{\text{eff}} \simeq 230 \text{ K}$.
- ▶ This high contrast is *the* primary challenge for detecting radiation from exoplanets.

Thermal emission flux ratio

On the illuminated side, equate the planet's thermal emission with the absorbed stellar radiation — the energy balance of last lecture with the redistribution factor f , $4\pi R_p^2 F_{S,p}/f = (1 - A_B) F_{S,*} (R_*/a)^2 \pi R_p^2$ — to get the *total* flux ratio

$$\frac{F_{\oplus,p}}{F_{\oplus,*}} = \frac{F_{S,p} R_p^2}{F_{S,*} R_*^2} = \left(\frac{R_p}{a} \right)^2 \frac{f}{4} (1 - A_B)$$

- ▶ Here f is the correction factor to 4π that describes the unknown redistribution of the absorbed stellar radiation, and the fluxes are total fluxes, integrated over all frequencies.
- ▶ The ratio depends on R_p/a , not on R_p/R_* : the planet reradiates the fraction $\pi R_p^2/4\pi a^2$ of the stellar luminosity it intercepts, times $(1 - A_B)$, times the redistribution factor.
- ▶ For a hot Jupiter at $a = 0.05$ AU with $A_B = 0.1$ and $f = 1$: $(R_J/a)^2 \times 0.9/4 \simeq 2 \times 10^{-5}$ bolometrically. Yet the observed ratio in an infrared band is $\sim 10^{-3}$, because the planet's emission is concentrated at long wavelengths where the star is faint — which is why we need a frequency-dependent estimate.

Thermal emission flux ratio at a given frequency

A more useful planet-star flux ratio estimate is **frequency dependent**, because telescope observations are limited to a fixed frequency or wavelength range. Approximate the star and planet surface fluxes by blackbody fluxes $\pi B(T, \nu)$, with T the effective temperature of the star and the equilibrium temperature of the planet:

$$\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{F_{S,p}(\nu)R_p^2}{F_{S,*}(\nu)R_*^2} = \frac{e^{h\nu/k_B T_{\text{eff},*}} - 1}{e^{h\nu/k_B T_{\text{eq},p}} - 1} \frac{R_p^2}{R_*^2}$$

- ▶ The ν^3 prefactors of the two Planck functions cancel; only the exponential factors and the area ratio remain.
- ▶ In the Wien tail of the planet ($h\nu \gg k_B T_{\text{eq}}$) the ratio is $\simeq e^{-h\nu(1/k_B T_{\text{eq}} - 1/k_B T_{\text{eff},*})} (R_p/R_*)^2$, exponentially small; it rises steeply toward the infrared and saturates in the Rayleigh–Jeans regime (next slide).
- ▶ Note that it is now R_p/R_* that matters, not R_p/a : at a fixed planet temperature, a smaller star gives a larger contrast. This is the same “M-dwarf opportunity” as for transits.
- ▶ Example: HD 189733b at $8\text{ }\mu\text{m}$ with $R_p/R_* = 0.155$, $T_{\text{eq}} = 1200\text{ K}$, $T_{\text{eff},*} = 5000\text{ K}$ gives $(e^{3.6} - 1)/(e^{1.5} - 1) \times 0.024 = 0.0034$, close to the observed eclipse depth of 0.34% (Lecture 1).

The Rayleigh–Jeans limit

At mid-infrared wavelengths a further simplification applies: for reasonable planet and star temperatures both blackbody fluxes are in the **Rayleigh–Jeans** region, $h\nu/k_B T \ll 1$. With $e^x - 1 \simeq x$ the Planck function becomes

$$\pi B(T, \nu) \simeq \pi \frac{2\nu^2}{c^2} k_B T,$$

linear in T , and the flux ratio becomes

$$\boxed{\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{F_{S,p}(\nu)R_p^2}{F_{S,*}(\nu)R_*^2} \simeq \frac{T_{\text{eq},p}}{T_{\text{eff},*}} \left(\frac{R_p}{R_*}\right)^2}$$

in the frequency range where the approximation is valid.

- ▶ This is the *maximum* contrast the thermal emission can reach: the temperature ratio times the area ratio.
- ▶ **Hot Jupiter:** $(1500/5800) \times (0.1)^2 \simeq 2.6 \times 10^{-3}$, approached already at $\lambda \gtrsim 10 \mu\text{m}$ (for $T_p = 1500 \text{ K}$, $h\nu/k_B T = 1$ at $\lambda = 9.6 \mu\text{m}$).
- ▶ **Earth:** $(255/5800) \times (0.0092)^2 \simeq 3.7 \times 10^{-6}$, reached only at $\lambda \gtrsim 100 \mu\text{m}$; at $10 \mu\text{m}$ the actual ratio is $\sim 10^{-7}$ because Earth is still on the Wien side.
- ▶ The Rayleigh–Jeans form makes explicit why the infrared is the place to detect thermal emission: the contrast saturates at T_p/T_* instead of decaying exponentially.

Using brightness temperatures instead of blackbodies

Real planets and stars deviate from blackbodies, sometimes strongly (the M-dwarf spectrum of last lecture, molecular bands in the planet). The honest version of the Rayleigh–Jeans flux ratio uses the **brightness temperatures** at each frequency:

$$\boxed{\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{F_{S,p}(\nu)R_p^2}{F_{S,*}(\nu)R_*^2} = \frac{T_{b,p}(\nu)}{T_{b,*}(\nu)} \left(\frac{R_p}{R_*}\right)^2} \quad (\text{Rayleigh–Jeans regime}),$$

and, outside the Rayleigh–Jeans regime, the ratio of Planck functions evaluated at $T_{b,p}(\nu)$ and $T_{b,*}(\nu)$.

- ▶ This is exactly how a measured **secondary-eclipse depth** is turned into a dayside brightness temperature: with R_p/R_* from the transit and a stellar model for $T_{b,*}(\nu)$, invert the Planck function for $T_{b,p}(\nu)$.
- ▶ The stellar brightness temperature matters: at $3.6\text{--}8\,\mu\text{m}$ a K or M dwarf has $T_{b,*}$ well below its T_{eff} because of molecular absorption, which raises the apparent planet-to-star contrast.
- ▶ $T_{b,p}(\nu)$ across several bands maps out the emission spectrum: bands with lower T_b are absorption features formed high in a (normally stratified) atmosphere; bands with higher T_b indicate a temperature inversion.

Scattered-light flux ratio at full phase

Scattered starlight is typically at visible wavelengths (extending to longer wavelengths for colder planets). At $\alpha = 0$, start again from the thermal flux ratio and replace the absorbed-and-reradiated factor by a scattered one:

$$\frac{f}{4}(1 - A_B) \longrightarrow \frac{g}{4}A_S(\nu),$$

where $A_S(\nu)$ is the spherical albedo (we want a wavelength-dependent ratio; monochromatic scattering assumed) and g , like f , is a correction to 4π : the solid angle into which the incident radiation is scattered compared to 4π . Scattered light cannot be advected around the planet; it leaves the illuminated hemisphere, with a directional pattern. Thus

$$\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{R_p^2}{a^2} \frac{g}{4} A_S(\nu).$$

The critical step is g : the ratio of energy emitted at zero phase angle to that emitted at all phase angles. With E_{scat} from the spherical-albedo derivation and Ψ_α the fraction scattered toward the observer,

$$\frac{g}{4\pi} = \frac{E_{\text{scat},\alpha=0}}{E_{\text{scat}}} = \frac{\Psi_{\alpha=0}(\nu)}{2\pi \int_0^\pi \Psi_\alpha(\nu) \sin \alpha d\alpha} = \frac{1}{\pi q(\nu)} = \frac{A_g(\nu)}{\pi A_S(\nu)} \implies \boxed{g = \frac{4A_g(\nu)}{A_S(\nu)}}$$

using $\Psi_{\alpha=0} = A_g/\pi$, the phase function Φ_α , and the phase integral $q = A_S/A_g$ (last lecture).

Scattered-light flux ratio: the result and numbers

Substituting g , the spherical albedo cancels and the full-phase visible flux ratio is

$$\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = A_g(\nu) \frac{R_p^2}{a^2}$$

- ▶ A common misunderstanding: the visible flux ratio at full phase involves the *geometric* albedo, which describes the light scattered toward the observer at $\alpha = 0$, not the spherical or Bond albedo, which describe the energy scattered in all directions. That is precisely what g converts.
- ▶ The ratio falls as a^{-2} and does not involve R_* at all (for a fixed stellar flux at the planet, a bigger star only means a bigger a for the same temperature).
- ▶ **Jupiter:** $A_g(500\text{nm}) = 0.5$, $a = 5.2\text{ AU}$, $R_p = R_J$ gives $F_{\oplus,p}/F_{\oplus,*} = 4.4 \times 10^{-9}$.
- ▶ **Scaled to 0.05 AU:** 10^4 times higher, 4.4×10^{-4} . The actual hot Jupiters are darker: Jupiter is bright because of its icy NH_3 and H_2O clouds, whereas hot Jupiters lack bright clouds and have strong Na/K absorption, so $A_g \lesssim 0.1$ and the ratio is $\lesssim 10^{-4}$ — consistent with the Kepler and MOST upper limits and detections.
- ▶ **Earth:** $A_g(500\text{--}800\text{nm}) = 0.2$, $a = 1\text{ AU}$: 2.1×10^{-10} at $\alpha = 0$ (the number quoted in Lecture 1).

Transmission flux ratio

During transit, starlight passes through the planet atmosphere and picks up its spectral features: the **transmission spectrum**. What is measured is the total (star + planet) flux in transit divided by the out-of-transit flux. Let R_p be the radius where the planet is opaque at all wavelengths of interest and A_H the additional radial extent of the atmosphere. Splitting the stellar disk into the unobscured part, the opaque planet ($I_p = 0$), and the atmospheric annulus:

$$F_{\text{in trans}, \oplus}(\nu) = \int_0^{2\pi} \int_{(R_p + A_H)/D_{\oplus}}^{R_*/D_{\oplus}} I_* \cos \omega \sin \omega d\omega d\phi + \int_0^{2\pi} \int_{R_p/D_{\oplus}}^{(R_p + A_H)/D_{\oplus}} I_{\text{atm}} \cos \omega \sin \omega d\omega d\phi.$$

Dividing by the out-of-transit stellar flux $F_{*, \oplus} = F_{S, *} (R_*/D_{\oplus})^2$ and assuming $A_H \ll R_p$,

$$\boxed{\frac{F_{\text{in trans}, \oplus}(\nu)}{F_{*, \oplus}(\nu)} = 1 - \left(\frac{R_p}{R_*}\right)^2 - \frac{2R_p A_H}{R_*^2} \left[1 - \frac{F_{S, p, \text{trans}}(\nu)}{F_{S, *, \text{trans}}(\nu)}\right]}$$

- First term: the star; second: the opaque disk (the transit depth); third: the annulus $2\pi R_p A_H$ over the stellar disk, times the fraction of starlight the annulus *blocks*. $F_{S, p, \text{trans}}/F_{S, *, \text{trans}}$ is the transmitted fraction, computed in Chapter 6 from the slant optical depth.

How large is the transmission signal?

The maximum signal occurs where the annulus is completely opaque ($F_{S,p,\text{trans}} = 0$):

$$\frac{F_{p,\oplus,\text{trans}}(\nu)}{F_{*,\oplus}(\nu)} = \frac{2R_p A_H}{R_*^2} \left[1 - \frac{F_{S,p,\text{trans}}}{F_{S,*,\text{trans}}} \right] \approx \frac{2R_p A_H}{R_*^2}$$

- ▶ The choice of A_H does not matter as long as it is large enough: above some height there is no atmosphere left to attenuate, the transmitted fraction is 1, and the bracket vanishes. What matters is the height at which the slant path becomes opaque, a few scale heights above the opaque radius.
- ▶ **Hot Jupiter** around a Sun-sized star: an H/He atmosphere with $T_{\text{eq}} \simeq 1000$ K has a scale height $H \simeq 500$ km (Chapter 9). Taking the atmosphere to extend $A_H = 5H$:
 $2R_p A_H / R_*^2 = 2 \times (7.1 \times 10^4) \times (2.5 \times 10^3) / (7 \times 10^5)^2 \simeq 7 \times 10^{-4}$, i.e. of order 10^{-3} (Seager quotes 1.5×10^{-3}).
- ▶ One part per thousand may seem very small. In fact it is a *large* number in the field of exoplanet atmosphere detection: compare with the $\lesssim 10^{-4}$ reflected-light and $\sim 10^{-3}$ thermal ratios above, and note that it requires no separation of the planet's light from the star's, only precise photometry in and out of transit.

Transmission signals of cooler planets, and the three channels

- ▶ **Jupiter analog** (a Jupiter-sized planet in a Jupiter-like orbit, $T_{\text{eq}} \simeq 120 \text{ K}$, $H = 24 \text{ km}$): the estimated transmission strength is only $\sim 4\text{--}7 \times 10^{-5}$. Cold planets have small scale heights and thus weak transmission spectra, even though their transit depth is the same.
- ▶ **Earth twin** ($H = 8 \text{ km}$, N_2 atmosphere): $\sim 2 \times 10^{-7}$ (Lecture 1). This is why terrestrial-planet transmission spectroscopy targets M dwarfs, where R_*^2 is 25–100 times smaller.
- ▶ The transmission technique has been very successful for hot Jupiters around bright stars: Na, K, H_2O , CH_4 , CO, CO_2 , He, and escaping H have all been identified this way, and JWST now routinely delivers transmission spectra of sub-Neptunes.

The three observational channels and what controls each:

channel	flux ratio	key planet property	geometric factor
thermal emission	$(T_{\text{b},p}/T_{\text{b},*})(R_p/R_*)^2$	temperature $T_{\text{b}}(\nu)$	$(R_p/R_*)^2$
reflected light	$A_g R_p^2/a^2$	geometric albedo $A_g(\nu)$	$(R_p/a)^2$
transmission	$2R_p A_H/R_*^2$	scale height H	R_p/R_*^2

Each channel favors different planets: emission and transmission favor hot, large planets around small stars; reflection favors close-in planets but is limited by their low albedos.

Visible-wavelength phase curves

An exoplanet goes through illumination phases like the Moon. We want the flux ratio not only at $\alpha = 0$ but as a function of α . Recall the normalized phase function

$$\Phi_{\alpha}(v) = \frac{\Psi_{\alpha}(v)}{\Psi_{\alpha=0}(v)}, \quad \Psi_{\alpha}(v) = \int_{\alpha-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{\text{inc}}(\theta', \phi', v) p \cos(\alpha - \phi) \cos \phi \cos^3 \theta \, d\theta \, d\phi.$$

The Lambert sphere is the one case with an analytic phase curve. With $p = 1$ and uniform incident intensity, the θ integral gives $4/3$ and the ϕ integral gives $\int_{\alpha-\pi/2}^{\pi/2} \cos(\alpha - \phi) \cos \phi \, d\phi = \frac{1}{2} [\sin \alpha + (\pi - \alpha) \cos \alpha]$, so after normalizing to $\alpha = 0$ (where the integral is $\pi/2$),

$$\Phi_{\alpha} = \frac{1}{\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha]$$

frequency independent, with $\Phi_0 = 1$, $\Phi_{\pi/2} = 1/\pi$, $\Phi_{\pi} = 0$.

- ▶ At quadrature ($\alpha = 90^\circ$) a Lambert sphere is $1/\pi \simeq 0.32$ as bright as at full phase, *less than half* (exercise 3.7): the visible illuminated half-disk is lit obliquely near the terminator and its brightest region (the substellar point) sits at the limb, seen at grazing angle.
- ▶ The phase integral follows: $q = 2 \int_0^{\pi} \Phi_{\alpha} \sin \alpha \, d\alpha = 3/2$.

Phase curves of Solar System bodies

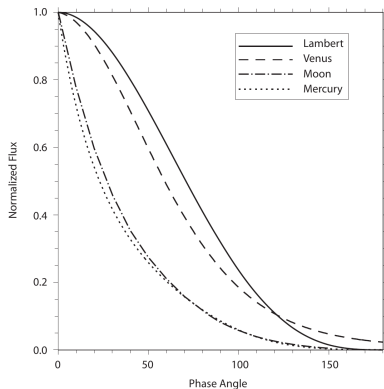


Figure 11: Visible-wavelength phase curves of Venus, Mercury, and the Moon, with the Lambert sphere (rightmost solid line). From formulae in Cox (2000); reproduced from Seager (2010), Fig. 3.9.

- ▶ **Venus**, with its thick cloud deck, follows the Lambert curve closely out to $\alpha \sim 100^\circ$ and then falls faster (forward scattering by cloud droplets brightens it again at very large α , not shown).
- ▶ The **Moon** and **Mercury** — airless, rough, regolith-covered — fall off much faster: their surfaces backscatter (the *opposition surge* near $\alpha = 0$) and shadowing in the rough surface darkens the oblique views. At quadrature the Moon is only ~ 0.1 of its full-phase brightness, not 0.32.
- ▶ Lesson: the phase curve shape encodes the *scattering physics* (clouds versus bare surface, particle size, surface roughness). For exoplanets it is a second observable beyond A_g .

Flux ratio at all phases and an Earth example

The normalized phase curve combined with the full-phase flux ratio gives the planet-star flux ratio at any phase:

$$\frac{F_{p,\oplus,\alpha}(\nu)}{F_{*,\oplus}(\nu)} = A_g(\nu) \frac{R_p^2}{a^2} \Phi_\alpha(\nu)$$

Earth orbiting the Sun as an exoplanet.

- ▶ At $\alpha = 0$ with $A_g(500\text{--}800\text{nm}) = 0.2$, $a = 1\text{ AU}$, $R_p = R_\oplus$: 2.1×10^{-10} . Earth is almost ten billion times fainter than the Sun — and $\alpha = 0$ is when the planet is directly behind the star.
- ▶ At **maximum elongation** ($\alpha = 90^\circ$), the natural configuration for direct imaging, approximating Earth as a Lambert sphere: $\Phi_{90^\circ} = 1/\pi$ and the ratio is 1.2×10^{-10} , a factor π lower. (Earth's real phase curve is not Lambertian: ocean glint and forward-scattering clouds modify it.)
- ▶ For an edge-on orbit the full range $0 < \alpha < 180^\circ$ is swept once per orbit; the amplitude of the reflected-light phase curve is $A_g R_p^2 / a^2$ and its *shape* constrains Φ_α , hence q and A_g . For inclination i , the accessible range is $90^\circ - i \leq \alpha \leq 90^\circ + i$ (exercise 3.8).

Infrared phase curves

Phase variation also occurs at *infrared* wavelengths, from thermal emission, and not just at visible wavelengths from scattered starlight. Thermal phase variation arises when the emission is dominated by reemitted absorbed starlight rather than by internal energy.

- ▶ The prime examples are the **tidally locked** (synchronously rotating) hot Jupiters, hot Neptunes, and hot Earths with $a < 0.05$ AU, strongly heated by their stars. They present the same face to the star at all times, just as the Moon does to Earth.
- ▶ The permanent day and night sides set up a temperature gradient that atmospheric circulation may or may not even out. How the absorbed stellar energy is circulated is a complex topic (Chapter 10).

Simplest case: a hot tidally locked planet where the incident stellar energy is absorbed in one layer, instantaneously heats that surface element, and is reemitted, with no advection. Thermal emission is isotropic, so each element reradiates like a Lambert scatterer and the problem is solved in analogy with the Lambert sphere: the normalized thermal phase curve is again

$$\Phi_{\alpha} = \frac{1}{\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha],$$

peaking at secondary eclipse and vanishing at transit (no nightside emission at all), with the dayside temperature corresponding to $f' = 2/3$.

Infrared phase curves: the general case

Beyond the simplest case, if the thermal flux distribution over the planet's surface is available from a model atmosphere or a circulation model, the thermal phase curve is still computed from the planet-star flux ratio, with the phase-dependent planetary flux

$$F_{\alpha,\text{thermal}}(\nu) = \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{\alpha,\text{thermal}}(\theta, \phi, \nu) \cos \phi \cos^2 \theta \, d\theta \, d\phi,$$

integrated over the hemisphere facing the observer at phase α (the subscript “thermal” denotes the thermal-infrared component of the intensity), and the flux ratio follows with R_p^2/R_*^2 and the stellar flux at that frequency.

- ▶ The shape of $I_{\alpha,\text{thermal}}(\theta, \phi)$ encodes the longitudinal temperature map: a hot spot shifted east of the substellar point produces a phase-curve peak *before* secondary eclipse (Lecture 1).
- ▶ The nightside flux sets the minimum of the phase curve near transit; the day–night amplitude measures how efficiently heat is redistributed, i.e. the effective f' .
- ▶ Real hot Jupiters (HD 189733b, HD 209458b) show amplitudes reduced relative to the no-redistribution Lambert curve and offset peaks: heat is carried to the nightside by winds. Ultra-hot Jupiters and planets on eccentric orbits (HD 80606b) show the full range from strong to weak redistribution.

Take-home messages

Temperatures. Energy balance gives $T_{\text{eq}} = T_{\text{eff},*}(R_*/a)^{1/2}[f'(1 - A_B)]^{1/4}$, with $f' = 1/4$ for uniform redistribution and $2/3$ for an instantaneously reradiating dayside. T_{eff} matches the total emitted flux and exceeds T_{eq} when the interior supplies heat; $T_b(\nu)$ matches the flux at one frequency and is the only temperature measurable for an exoplanet.

Albedos. A_g compares the full-phase flux with a Lambert disk (Lambert sphere: $2/3$; A_g can exceed 1); $A_S = A_g q$ with the phase integral q ($3/2$ for Lambert); A_B is A_S averaged over the stellar spectrum and is the one that enters the energy balance.

Flux ratios. $F_{\oplus,p}/F_{\oplus,*} = F_{S,p}R_p^2/F_{S,*}R_*^2$. Thermal: $(R_p/a)^2(f/4)(1 - A_B)$ bolometrically, $(T_{b,p}/T_{b,*})(R_p/R_*)^2$ in the Rayleigh–Jeans regime. Reflected at full phase: $A_g R_p^2/a^2$ (geometric, not Bond, albedo; $g = 4A_g/A_S$). Transmission: $2R_p A_H/R_*^2 \sim 10^{-3}$ for hot Jupiters.

Phase curves. $A_g(R_p/a)^2\Phi_\alpha$ with the Lambert phase function $\Phi_\alpha = [\sin \alpha + (\pi - \alpha) \cos \alpha]/\pi$; thermal phase curves follow the same law when the absorbed energy is reradiated instantaneously, and depart from it when winds redistribute heat.

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